Introduction to Machine Learning

Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data.

Supervised learning uses labeled training examples to teach models how to make predictions on new data.

Neural networks consist of layers of interconnected nodes that transform input features into useful representations.

Deep learning uses many layers to capture complex patterns in images, text, and speech.

Evaluation metrics such as accuracy, precision, and recall help researchers measure model quality.

Cross-validation reduces overfitting by testing models on unseen portions of the dataset.

Feature engineering and data cleaning remain essential before training because noisy data can mislead algorithms.

Gradient descent optimizes model parameters by iteratively reducing prediction error on training batches.